

claude-windows / 45ea3a91-654d-4972-9cd3-65f35db54caf

Metadata

```
- Source: `claude-windows`  
- Kind: `claude`  
- Source file: `C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\45ea3a91-654d-4972-9cd3-65f35db54caf\subagents\workflows\wf_f10e69b6-333\agent-ac663e5ed28c91b94.jsonl`  
- SHA-256: `56b11579c5160dca457f51ef1c62446d209631a088115bb33f22d7a2e6ad2005`  
- Source modified: `2026-06-15T14:59:21+00:00`  
- Imported at: `2026-07-05T16:48:27+00:00`  
- project: `wf_f10e69b6-333`  
- session_id: `45ea3a91-654d-4972-9cd3-65f35db54caf`
```

Transcript

1. user (2026-06-15T14:57:50.096Z)

You are reviewing a DRAFT product-overview document for "KhelSutra" ? a local, AI badminton highlight/rally tool.
The draft lives at C:/Users/avidu/Projects/badminton-highlight-indexer/docs/KHELSUTRA_PRODUCT_OVERVIEW.md (Read it). It is dual-audience: (a) a non-technical FIELD ASSOCIATE who will talk to academies, and (b) invited ALPHA/BETA academies/coaches/players who will use the tool and give feedback. It must be PRODUCT-LEVEL (no fine-grained technical detail) and HONEST (no overpromising to real prospective customers).

GROUND TRUTH (from the repo's docs/memory ? what is actually real):

- TODAY the tool genuinely does: upload a long video ? automatic rally detection ? highlight reel (with a "Generated by KhelSutra.guru" watermark) ? a simple summary (rally count + total play time); re-process/download;
private local-first web app (async jobs + chunked upload + highlight download are live & E2E-verified); runs on real 4K GoPro match footage.
- Quality reality (DO NOT put raw numbers in a customer doc): clean single-court footage works very well (~0.93 F1);
busy MULTI-COURT footage is markedly harder (~0.66) and is the explicit moat/focus. The doc should convey this qualitatively ("very good on clean single-court; multi-court is harder and is what we're improving").
- ROADMAP (NOT today): shot detection (smash/drop/serve), deeper per-rally/per-match analysis, per-player history,
"AI assistant coach", more sports (table tennis next), Kannada/Hindi localization.
- THE FLYWHEEL (the headline the owner wants foregrounded): players/coaches can run their OWN videos; coaches can
correct what it found; over time + more videos the model sharpens AND that library unlocks the deeper analysis.
- GUARDRAILS (hard, non-negotiable, must be represented accurately): using a user's video to improve/train the model
is OPT-IN (explicit permission), never automatic; footage involving MINORS is NEVER used for model improvement;
private/local-first ("your video, your data"); pricing is NOT set.
- Capture quality is the gate: fixed camera behind court, full court in frame, landscape, steady, decent light;
handheld/panning/broadcast footage is the failure mode.
- It is NOT line-calling/scoring/umpiring, NOT player biometrics, NOT a coach replacement.

RULES: read-only; you cannot edit. Return STRUCTURED output (consumed programmatically). Quote the draft line/phrase when flagging. Be specific and actionable. Do not invent product capabilities.

LENS: HONESTY / OVERPROMISE GUARD. Your job: catch any sentence a real alpha customer could read as a promise the product does not keep TODAY. Flag anything that blurs today-vs-roadmap, implies the AI-coach/shot-detection exists now, implies accuracy is perfect/guaranteed, or implies multi-court is solved. Verify the "Today" section claims ONLY what ground truth lists as real today. Verify the roadmap section is unambiguously framed as direction, not current features. The doc must under-promise and let the demo over-deliver.

2. assistant (2026-06-15T14:57:53.540Z)

I'll read the draft document and the ground truth from the repo to review it through the honesty/overpromise lens.

3. user (2026-06-15T14:57:54.517Z)

```
1      # KhelSutra ? Product Overview & Early-User Guide
2
3      *What it does today, how to get the best out of it, and where it's going.*
4      *For field associates and for invited alpha / beta academies, coaches, and players.*
5
6      ---
7
8      ## In one line
9
10     **KhelSutra turns ordinary badminton video into useful pieces ? automatically.** Point a
phone or
11     GoPro at the court, record a session, and KhelSutra finds every rally, stitches a
**highlight reel**,
12     and gives you a **simple summary** (how many rallies, how much real play time). No
special hardware,
13     no editing, no manual scrubbing.
14
15     ---
16
17     ## The problem we're solving
18
19     Academies and serious players already record a lot of video ? and then almost never
watch it. Hours
20     of footage pile up on phones and memory cards because cutting it into useful clips by
hand is tedious
21     and nobody has the time. The good rallies, the moments worth reviewing, stay buried.
22
23     And most academy halls run **several courts at once**, so a single recording is a busy,
noisy scene.
24     Ordinary video tools choke on that. **Pulling the real rallies out of that footage ?
automatically ?
25     is exactly what KhelSutra is built to do.**
26
27     ---
28
29     ## Who it's for
30
31     - **Coaching academies & coaches** ? turn raw session footage into review-ready clips
and reels for
32     teaching, without an editor.
33     - **Serious players** ? get clips of your own rallies to review, keep, and share.
34     - Today's focus is **badminton academies in Bengaluru** ? that's who we're building with
first.
35
36     ---
37
38     ## What it does **today**
39
```

40 This is what's working right now, in the private web app invited users can try:

41

42 - ****Upload a full session or match video**** (from a phone on a tripod, or a GoPro). Long videos are fine.

43 - ****Automatic rally detection**** ? it finds where each rally starts and ends, on its own.

44 - ****Highlight reel**** ? the rallies stitched into one clip you can watch and share, with a small

45 **"Generated by KhelSutra.guru"** mark.

46 - ****A simple summary**** ? how many rallies were found and how much actual play time there was.

47 - ****Re-use & download**** ? process once, come back to it, download the reel.

48

49 That's the core loop: ****record ? upload ? get back a reel + summary.**** No editing skill needed.

50

51 ---

52

53 **## How to get the best out of it (capture matters)**

54

55 KhelSutra works best on a ****steady, full-court recording**** ? the same way a coach would set a camera

56 to review play. A few simple habits make a big difference to the results:

57

58 - ****Put the camera behind the court, slightly elevated****, with the ****whole court in frame****.

59 - ****Keep it still**** ? a fixed tripod beats a handheld or panning shot every time.

60 - ****Shoot landscape****, with reasonable lighting.

61 - ****Record the whole session**** and upload it as-is ? KhelSutra does the cutting.

62

63 Footage that fights these habits (shaky handheld, heavy zooming/panning, broadcast-style cuts, very

64 dark halls) is the ****hardest case**** and where results suffer most. When in doubt: fixed camera, full

65 court, don't touch it.

66

67 > A one-page capture checklist is provided to academies in the trial.

68

69 ---

70

71 **## The part that makes KhelSutra different: **it learns from your videos****

72

73 This is the heart of the product, and what sets it apart from a generic "AI clipper":

74

75 ****Your videos make _your_ tool sharper.**** As you and your players run more of your own footage through

76 KhelSutra ? and as coaches confirm or correct what it found ? the underlying model gets better at

77 ***your* courts, *your* lighting, *your* players, and the busy multi-court halls that trip up everything**

78 else. It improves with use instead of staying frozen.

79

80 That same growing library of real, labelled footage is also what unlocks the ****deeper features below**** ?

81 which is why early users aren't just customers, they're co-builders. Every session you contribute makes

82 the next version smarter.

83

84 ****Your data stays yours.**** Using your video to improve the model is ****opt-in**** ? nothing is used to train

85 anything without your explicit permission. We don't publish anyone's footage. ****Sessions involving minors**

86 are never used for model improvement.** The product is built ****private and local-first****: your video, your

87 data, your control.

88

```

89     ---
90
91     ## Where it's going ? toward an **AI assistant coach**
92
93     Today KhelSutra finds rallies and makes reels. The roadmap ? fed by the growing video
94     library above ?
95     turns it into a genuine training companion:
96
97     - **Shot detection** ? recognising smashes, drops, net shots, serves, and clears within
98     each rally.
99     - **Deeper per-rally and per-match analysis** ? patterns, tendencies, what's working and
100     what isn't.
101     - **Per-player & per-match history** ? track a player's progress over weeks and months,
102     not just one clip.
103     - **Personalised insights** ? an **AI assistant coach** that supports (never replaces)
104     the real coach,
105     surfacing what to work on next.
106     - **More sports** ? table tennis next, then other racket/court sports.
107     - **Your language** ? Kannada and Hindi, alongside English.
108
109     > **Read this honestly:** the items in this section are the **direction**, not today's
110     buttons. What you
111     > can use *today* is rally detection + highlight reels + the play-time summary. The
112     deeper analysis arrives
113     > as the video library grows ? which is exactly why your footage and feedback matter so
114     much.
115
116     ---
117
118     ## What we're focused on over the next ~4 weeks
119
120     1. **Growing the library of real academy footage** ? especially busy **multi-court**
121     halls (the hard,
122     valuable case). This is the single biggest lever on quality.
123     2. **Field conversations with academies** ? understanding how coaches and players
124     actually work with
125     video, and what would make this genuinely useful.
126     3. **Recording sessions and showing academies their own footage** as reels ? "this is
127     what it looks
128     like for you."
129     4. **Sharpening multi-court accuracy** and inviting a few more academies into the
130     private trial.
131
132     ---
133
134     ## What KhelSutra is **not** (so expectations stay honest)
135
136     - It's an **early, invite-only trial** ? useful, but still improving; you'll hit rough
137     edges.
138     - It is **not a line-calling, scoring, or umpiring system** ? it finds rallies and
139     highlights.
140     - **Shot detection and the deeper "AI coach" analytics are on the roadmap, not available
141     today.**
142     - It is **not a replacement for a coach** ? it's a tool that saves coaches time and adds
143     insight.
144     - Accuracy depends on **capture quality** ; messy handheld/broadcast footage is a known
145     weak spot.
146     - **Pricing isn't set** ? we're still learning what's fair; today's invited use is about
147     getting it right.
148
149     ---
150
151     ## Quick answers to common questions
152
153     ***"How accurate is it?"***

```

136 Very good on a clean, single-court recording. A busy multi-court hall is harder ? and is exactly what
137 we're improving. The best answer is to *show* a sample reel rather than quote a number.
138
139 ***"Do I need a special camera or hardware?"**
140 No. A phone or GoPro on a tripod, behind the court, whole court in frame. We provide a one-page checklist.
141
142 ***"Is my video private? What happens to it?"**
143 It's built private and local-first ? your video stays yours. We never use anyone's footage to improve the
144 model without explicit opt-in, we don't publish footage, and **footage involving minors is never used**.
145
146 ***"Can it tell me the score / who won / call lines?"**
147 Not today. Today it finds rallies and builds a highlight reel + a play-time summary.
148
149 ***"Does it work for doubles and tournaments?"**
150 Yes for finding rallies. Crowded multi-court tournament halls are the hard case we're specifically working on.
151
152 ***"Will it get better?"**
153 Yes ? that's the whole idea. The more real footage runs through it (with permission) and the more coaches
154 correct it, the sharper it gets, and the sooner shot detection and deeper analysis arrive.
155
156 ***"Which sports?"**
157 Badminton today; table tennis and other racket sports are on the roadmap.
158
159 ***"When can I use it / how do I join?"**
160 It's an invite-only private trial. If you're interested, the best next step is to let us record a session
161 and show you your own court and players as a reel.
162
163 ---
164
165 *KhelSutra · khelsutra.guru · Internal & invited-trial overview, June 2026.*
166 *Field associates: the conversation script and visit rules live in the separate Field Visit Playbook.*
167

4. assistant (2026-06-15T14:58:02.013Z)

This is a read-only review task. I have the draft and the ground truth. Let me cross-check a couple of the live-capability claims against the repo's own docs to ground my honesty assessment, particularly the watermark text and what's actually E2E-verified.

5. user (2026-06-15T14:58:05.417Z)

```
config.json:79:          "text": "Generated by KhelSutra.guru",
docs\archives\decisions\DECISIONS.md:477:1. **Product name: KhelSutra Guru**; domain
**khelsutra.guru** (GoDaddy receipt: 3-yr
docs\KHELSUTRA_PRODUCT_OVERVIEW.md:45:  **"Generated by KhelSutra.guru" mark.
docs\KHELSUTRA_PRODUCT_OVERVIEW.md:165:*KhelSutra · khelsutra.guru · Internal & invited-trial
overview, June 2026.*
docs\HOSTING_PLAN.md:1:# Hosting Plan ? khelsutra.guru
docs\HOSTING_PLAN.md:27:  ??? app.khelsutra.guru ? Cloudflare Pages (React SPA; auto-deploy
from GitHub)
docs\HOSTING_PLAN.md:28:  ??? api.khelsutra.guru ? Cloudflare Tunnel (outbound-only
`cloudflared` service on the box)
docs\HOSTING_PLAN.md:49:browser ??? Cloudflare edge (free, unchanged) ??? app.khelsutra.guru
(Pages)
docs\HOSTING_PLAN.md:50:                                     ??? api.khelsutra.guru ???
Cloud Run (FastAPI control plane, scale-to-zero)
```

docs\HOSTING_PLAN.md:80:1. Create a free Cloudflare account ? **Add a domain** ?
`khelsutra.guru` ? Free plan. Cloudflare
docs\HOSTING_PLAN.md:82:2. GoDaddy ? **My Products ? khelsutra.guru ? Manage DNS ? Nameservers ?
Change ?
docs\HOSTING_PLAN.md:104:Access allowlist ON for both hostnames ? CORS narrowed to
`https://app.khelsutra.guru` ? rate
docs\NEXT_STEPS.md:5:ADR-009 ratified ? KhelSutra Guru / khelsutra.guru, private `khelsutra`
repo, local-first delivery ? see
docs\STORAGE_SHARING_MODEL.md:62:A Gmail-style **quota meter** is permanent UI. Profile =
`khelsutra.guru/@handle`: reel gallery,
docs\STORAGE_SHARING_MODEL.md:184:12. Profile namespace: `@handles` on khelsutra.guru?
docs\UI_ALPHA_EXECUTION_PLAN.md:12:| 2026-06-10 | Product name **KhelSutra Guru** . domain
khelsutra.guru (GoDaddy, 3-yr + protection) |
docs\UI_ALPHA_EXECUTION_PLAN.md:48:- [] **D1 (owner):** Cloudflare account ? add khelsutra.guru
? GoDaddy nameserver switch (HOSTING_PLAN \$5) ? scoped API token (\$6) ? Pages project connected
to GitHub `khelsutra` (build root `web/`)
docs\UI_ALPHA_EXECUTION_PLAN.md:49:- [] **D2 (Claude):** DNS records; `cloudflared` Windows
service on the box ? `api.khelsutra.guru`; Access apps + email-allowlist policies for `app` &
`api`; SPA prod env
docs\UI_PROPOSAL.md:1:# UI Proposal ? khelsutra.guru web app (alpha ? beta)
docs\UI_PROPOSAL.md:34:API = existing FastAPI on the GPU box via tunnel at `api.khelsutra.guru`.

[Showing results with pagination = limit: 20]

6. user (2026-06-15T14:58:05.538Z)

188:[Omitted long matching line]
214:[Omitted long matching line]

7. assistant (2026-06-15T14:58:09.621Z)

The watermark in `config.json` reads "Generated by Khelsutra.guru" (lowercase 's'), while the draft renders it as "Generated by KhelSutra.guru" (capital 'S'). Minor, but worth a note since the draft quotes it verbatim. Let me check what the actual summary/today capabilities are in CODE_MAP to confirm the "Today" claims, and check the current-state memory for E2E-verified scope.

8. user (2026-06-15T14:58:10.583Z)

180
181 ### 2.3a Audio (E1 probe; NOT adopted for fusion ? ADR-007)
182 - `@backend/pipeline/audio/hit\_detector.py` ? classical-onset shuttle-hit detector. Pure
numpy + stdlib `wave` (NO scipy); ffmpeg only for extraction. `::HitEvent`(t,strength),
`::extract\_audio` (ffmpeg -> mono 16k PCM WAV), `::load\_wav`, `::onset\_envelope` (pure-numpy
STFT spectral flux; `band=(lo,hi)` band-limit), `::detect\_hits` (adaptive `mean+k.std`;
`k`=sensitivity knob), `::detect\_hits\_in\_video`. ? separate signal ? WASB never modified by
this; per audio-e1-findings recall 100% but discrimination ~2.2x, band-pass null -> NOT adopted
(PR #35).
183 - `@backend/eval/audio/e1\_probe.py` ? E1 GO/NO-GO diagnostic (no training/fusion).
`::cluster\_hits` (groups thwacks; ? `min\_hits>=2` so a lone service-feed hit isn't a rally),
`::run` (discrimination ratio + recall recovery + audio-only P/R/F1), `::main`. reuses
`hit\_detector` + `calibrate\_wasb.{load\_golden\_gts,make\_predict\_fn,BASELINE}` + `metrics`.
184
185 ### 2.4 Stitcher / tools / utils
186 - `@backend/pipeline/stitcher/compiler.py::VideoCompiler` ? `compile_highlights(raw,
segments:[(start,end)], out_name)`. Per-clip re-encode -> `-f concat -c copy`. `::parse_time`
(mirrors gemini.parse_time; ? unparseable -> 0.0), `::get\_video\_duration`. ? if duration probe
fails (0.0) NO boundary validation; `begin\_padding`/`end\_padding` ctor defaults 0.5/1.0 but
every live call site (`/api/compile`, CLI trim, `scripts/run\_process\_and\_stitch.py`) threads
`config.json` stitching. `{begin\_padding,end\_padding}` (typed default 0.5/0.5); temp clips in a
TemporaryDirectory under output_dir; raises ValueError/FileNotFoundError/RuntimeError; uses
`print()`. ? two-stage codec contract: clips re-encoded with IDENTICAL settings precisely so
final concat can stream-copy ? change a preset and concat may break. ? optional **branding
watermark** (`stitching.watermark`, default-on): `::watermark\_filter` builds a `-vf` (drawtext

```

via `textfile=` + optional logo `movie/overlay`) applied to the per-clip re-encode only (concat
stays `-c copy`); a failed watermark encode retries the clip WITHOUT it (never nukes the reel,
and logs the ffmpeg stderr reason). Pass `watermark=` to `VideoCompiler(...)` (a
`WatermarkConfig` model or dict; `None` = off). Default font = bundled Apache-2.0
`::_BUNDLED_FONT` (`backend/assets/fonts/RobotoSlab-Regular.ttf`) so it renders without
fontconfig; `font_path` overrides.
187 - `@backend/tools/rally_slicer.py` ? standalone CLI. `::compute_clip_windows` (pure,
unit-tested), `::load_rally_labels` (golden schema; ? silently skips degenerate rows),
`::slice_rallies`, `::ClipWindow` (rally:str), `::BEGIN_PADDING=0.5`/`::END_PADDING=1.0` (?
duplicate compiler's by copy). `_read_time` -> backend/eval/annotations.parse_timestamp`.
188 - `@tools/cleanup_caches.py` ? **top-level** `tools/` ops tool (NOT backend runtime).
Dry-run-first cache GC. Pure `::classify` (name,is_dir,location)/`::plan_deletions` (unit-tested
= the destructive decision path) + I/O `::scan_wsl` (du+find; ? arg-globbing NOT a `for` loop ?
the login shell empties loop vars) / `::scan_windows` / `::delete`. CLI: `python -m
tools.cleanup_caches` (report) . `--apply` / `--keep-video <id|stem>` / `--older-than N` /
`--include-wasbcache` / `--summary` (hook nudge). Purges `*_frames`/staged/proxy/handoff/tmp;
keeps CSV/DB/reels. Companion to the runner self-clean (2.2). See `[[cache-hygiene-policy]]`
memory.
189 - `@backend/utils/video.py` ? `::get_video_duration` (ffprobe; ? 0.0 on failure =
"unknown"), `::extract_video_chunk(...,reencode=False,scale_width=None)`. ? copy mode cuts at
nearest keyframe (drift) ? pass `reencode=True` for short/accurate clips; `scale_width` requires
`reencode=True`.
190 - `@backend/utils/geometry.py` ? `::get_velocity_normalised` (fraction of frame-width/s;
? raw-px fallback if width<=0; used by yolo_hybrid), `::calculate_iou` (? +1 px convention
inflates IoU), `::get_center/get_distance/get_velocity`.
191
192 ### 2.5 Validators ?
193 - `@backend/pipeline/validators/base.py::VideoValidator` ABC, `::ValidationResult`
(pydantic: validator_name,passed,message,details), `::ValidationRegistry` (ordered LIST ?
registration order = exec order; ? no dedup), `::registry` (singleton). § `validate(video_path,
config, context)` -> `ValidationResult`. `::run_all_with_context` -> (results, context) (wraps
each validate in try/except -> failed result not abort); `::run_all` discards context. ? **add a
validator**: subclass + `registry.register(MyValidator())`.
194 - `@backend/pipeline/validators/__init__.py` ? registers in EXEC order: FormatValidator,
ResolutionFPSValidator, PTSContinuityValidator, SceneCutDensityValidator, BlurValidator,
RelevanceValidator (? producers before consumers ? format_check first; ? import mutates global
registry as side effect).
195 - `@backend/pipeline/validators/format.py::FormatValidator` (`format_check`) ?
**PRODUCER** of `context['ffprobe_meta']`; ? MUST run before resolution_fps; container check is
substring `mp4` (so MOV-family passes); external `ffprobe`.
196 - `@backend/pipeline/validators/resolution_fps.py::ResolutionFPSValidator`
(`resolution_fps_check`) ? CONSUMES `ffprobe_meta` (? hard-fails "Run format_check first" if
absent). `min_fps` 29.9; min res 1280x720.
197 - `@backend/pipeline/validators/pts_continuity.py::PTSContinuityValidator`
(`pts_continuity_check`) ? re-probes (no context). ? 10.0s keyframe-gap threshold hardcoded
(comment says 8.0s ? drift, NOT config-driven); any backward jump fails; `<2` keyframes PASSES
(too short).
198 - `@backend/pipeline/validators/scene_cut.py::SceneCutDensityValidator`
(`scene_cut_check`) ? OpenCV. ? `cut_threshold=0.6` hardcoded (NOT config); samples ~2fps, caps
at 100 SAMPLED frames (~first 50s); `max_scene_cuts_per_minute` (0.5) is the config fail
threshold.
199 - `@backend/pipeline/validators/blur.py::BlurValidator` (`blur_check`) ? Laplacian var <
`min_blur_threshold` (default 100.0 in-file, 20.0 in models/config) = fail; samples 15 frames. ?
magic absolute, not resolution-normalized.
200 - `@backend/pipeline/validators/relevance.py::RelevanceValidator` (`relevance_check`) ?
HSV court-green ratio (? purely color, no geometry despite name); lazy `import backend.sports`;
3-tier config precedence (`validation.relevance` -> sport profile -> hardcoded); ? unknown sport
-> empty cfg, falls back to defaults, does NOT fail.
201
202 ### 2.6 Sports / Providers / Annotations ?
203 - `@backend/sports/base.py::SportProfile` ABC
(`name`,`get_prompt`,`get_relevance_config`);
`@backend/sports/badminton.py::BadmintonSportProfile` (the Gemini prompt + HSV relevance cfg);
`@backend/sports/__init__.py::sport_registry` (+`::SportRegistry`, auto-registers badminton; ?
`register` instantiates profile to read `.name`). ? **add a sport**: subclass `SportProfile`,

```

```

`sport_registry.register(MyProfile)`. $ prompt emits JSON array `{start_time(float DECIMAL SEC),
end_time, shuttle_exchange_count(int), shot_count_confidence('Low'|'Medium'|'High')}`. ? prompt
actively defends against MM:SS; `get()` returns a fresh instance each call.
204 - `@backend/providers/base.py::AIVideoProvider` ABC (`__init__(api_key, model_name)`);
`@backend/providers/gemini.py::GeminiProvider` (lazy `::client` property; `::MAX_UPLOAD_MB=500`;
upload->poll->generate JSON temp 0->delete); `@backend/providers/__init__.py::provider_registry`
(`::ProviderRegistry`, auto-registers `gemini`). ? **add a provider**: subclass,
`provider_registry.register('name', MyCls)` (key passed EXPLICITLY); `get(name, api_key,
model_name)`. $ `analyze_video(path, prompt, chunk_duration=0.0, label='')` -> (segments:list,
metrics:dict{upload_time,process_time,gen_time}). ? blocking 10s poll; returns `([], metrics)`
on ANY failure (empty != error); oversize clip refused LOUDLY (`logger.error` +
`metrics['oversize_skipped']=True` ? callers aggregate into the report's `oversize_clip_skipped`
rule); orphaned remote file possible if process dies mid-run.
205 - `@backend/annotations/base.py::AnnotationProvider` ABC (`submit/poll/fetch/annotate`,
`::Interval` (= metrics.Interval), `::QUEUED/RUNNING/DONE/FAILED`/`::STATUSES`,
`::AnnotationProviderRegistry` (decorator-friendly `register` returns the class; ? keys off
`provider_cls().name`, falls back to `__name__` if no-arg ctor throws; `list_available()` ->
LIST not dict), `::annotation_registry`. ? **add a tier**: subclass (define `name`),
`annotation_registry.register(MyCls)` at module bottom + import in `__init__.py`. $ `fetch` ->
List[Interval]` sorted; contract says RAISE (not `[]`) on unavailable.
206 - `@backend/annotations/file_provider.py::FileProvider` (`'file'; job_id IS the
resolved CSV path; `_resolve` -> `

```

```

`finally:`); writes `<video>.report.json`/`.report.md`/`.summary.md` next to video.
`::OBSREPORT_AVAILABLE` ? soft-dep gate (True only if `import obsreport` AND `from . import
harvest` both succeed). `::report_enabled(config)` reads `config["report"]["enabled"]` default
True.
215 - `@backend/reporting/harvest.py` ? records raw FACTS into `obsreport.RunRecorder` (the
footgun RULES live in spec.yaml, not here). `::record_run(...)` ? top-level assembler (does NOT
write); builds RunRecorder, reads `config_default = config["indexing"]["default_segmenter"]`,
runs 4 stage builders + `_highlights`, `rec.finalize()``.
`::AI_SEGMENTERS={yolo_hybrid,tracknet_hybrid,wasb_hybrid,gemini}` ? (add new AI segmenters
here). `::video_meta_from_context` (derives VideoMeta from `val_context['ffprobe_meta']`; now
also `audio_present` for Input Quality reports extension; $ `fps_source` ?
{unknown,probed,fallback}), `::validation_stage`, `::segmentation_stage` ($ `details.{segment
er_requested,segmenter_actual,config_default,matches_config_default,segmenter_explicitly_request
ed,was_reingest,detector,metadata_source}`; ? marks `motion` output "synthetic/heuristic"),
`::ai_stage` ($ `details.{ai_based,provider,model,api_key_present,oversize_clips_skipped}`;
metric `confirmed_segments`), `::highlights` (coverage_pct). `record_run(run_signals=)` carries
`run_signals.pop(video_id)` counters into `_ai_stage`. ? `record_run` fps fallback:
`fps_source=='unknown'` + ran -> promotes to "fallback", forces fps=30.0 (mirrors runners).
`::SPEC_PATH`/`::load_spec` resolve `spec.yaml` next to module. (Soft import of obsreport so
pure helpers like video_meta are importable for tests/coverage even without the dep.)
216 - `@backend/reporting/spec.yaml` ? $ declarative report layout + footgun engine consumed
by obsreport. `::customer_verdict` (pass/pass_with_warnings/fail messages). `::sections`
(ordered `{id,title,kind,audience?{both,customer,technical}}`). ? `::rules` ? each
`{code,severity,when[{path,op,value}],message,faq_ref,customer_message?}`; `when` clauses AND'd;
`path` dotted-navigates the recorder JSON. ? `path` strings are a HARD coupling to harvest's
stage/metric/detail names ? renaming a detail in harvest.py silently breaks the matching rule.
Rules (9): `silent_provider_noop` (error: no api key + 0 segs), `oversize_clip_skipped` (error:
provider refused chunks over MAX_UPLOAD_MB ? 4K stream-copy footgun),
`ai_empty_vs_no_rallies` (warn), `fps_fallback_used` (warn), `audio_disabled_or_missing` (warn: no
audio stream -> loses the shuttle-thwack recall cue; reads `video_meta.audio_present`),
`synthetic_motion_metadata` (warn), `segmenter_divergence` (info),
`reingest_replaced_prior` (info), `validation_failed` (error: halts indexing).
217 - `@backend/reporting/run_signals.py` ? ? in-process, thread-safe channel pipeline-
internals -> report (`::incr(video_id,key,by)`, `::pop(video_id)`). No obsreport dep (always
importable). Segmenter worker threads record facts the harvest can't see (e.g.
`oversize_clips_skipped`); `emit_indexer_report` pops UNCONDITIONALLY (even reporting-off, so
nothing accumulates) and passes to `record_run`. ? process-local best-effort: signals are
dropped (by design) if no report is emitted for that video_id.
218
219 ---
220
221 ## 3. DATA CONTRACTS (canonical shapes ? one place)
222
223 $ **Trajectory CSV** (produced by tracknet/wasb predict, consumed by
`parse_trajectory_csv`):
224 `Frame,Visibility,X,Y` ? header+column order fixed; `Visibility==1`=visible; X,Y pixel
coords.

```

9. user (2026-06-15T14:58:11.400Z)

<system-reminder>[Truncated: PARTIAL view ? showing lines 1-450 of 1931 total (91083 tokens, cap 25000). Call Read with offset=451 limit=450 for the next page, or Grep to find a specific section. Do NOT answer from this page alone if the answer may be further in the file.]</system-reminder>

```

1 ---
2 name: current-state
3 description: "Live micro-checkpoint state for Claude ? updated every checkpoint (not
only on request). What just landed, what's next, open decisions."
4 metadata:
5   node_type: memory
6   type: project
7   originSessionId: b13673cc-f087-43b1-9527-35dc0b67b301
8 ---
9

```

```

10      # Current state ? updated continuously (see [[working-agreement-checkpoints]])
11
12      **Checkpoint:** 2026-06-15 (PRODUCT RAMP-UP DOC for the field associate + alpha/beta
users). **Owner found a field
13      associate** (willing to talk to coaches/serious players); associate's feedback = he
needs to understand the WHOLE TOOL
14      from a PRODUCT perspective before going out. Owner refined scope twice: (1) the doc
should ALSO serve invited alpha/beta
15      academies so they can use the tool well + give feedback; (2) **headline the flywheel** ?
players/coaches use THEIR OWN
16      videos to fine-tune the underlying model; over time + more videos it sharpens AND grows
into shot detection + deeper
17      skill analysis = an **AI Coach**. Building `docs/KHELSUTRA_PRODUCT_OVERVIEW.md`
(committable source) + a shareable PDF,
18      PRODUCT-LEVEL only (NO fine-grained tech: no model names/F1/pipeline internals). Honesty
rails: clearly separate TODAY
19      (rally detect + highlight reel + play-time summary, phone-on-tripod, private web app)
from ROADMAP (auto per-user
20      sharpening, shot detection, AI-coach analytics, multisport, Kannada); consent/opt-in +
minors-never-used + local-first
21      must be crisp. Complements (does NOT duplicate) the existing
`PRODUCT_FIELD_ASSOCIATE_FLYER.md` (recruiting) +
22      `FIELD_VISIT_PLAYBOOK.md` (visit script). Both repos synced to remote HEAD at session
start (indexer 5fe9beb/#176,
23      collector 031e410/#29) ? clean.
24
25      **Checkpoint:** 2026-06-15 (ROORA JUNE-15 BATCH + 3 collector PRs merged + guidelines
enriched). Owner-authorized
26      merging of "process" PRs (prep?verify?transcode?pre-ingest?ingest); billing outage still
red CI account-wide ?
27      `--admin` merged on local-green (precedent: indexer NEXT_STEPS). **Collector master
`031e410`:**
28      - **#27 preflight gate** (`src/core/batch_preflight.py`): prep-annotation-batch now
scans the tree FIRST ? BLOCKS on
29      duplicate videos (size+head+tail fingerprint, never full-hashes 10GB over USB),
surfaces empty dirs / VFR /
30      multi-video dirs; `--dry-run` (scan only) + `--allow-duplicates` (for the A/A test).
Header-only `proxy.probe_light`
31      (no `--count_packets`) ? 3.8s scan on the real 115GB tree. Caught the empty-TT-dir +
Video6/9-byte-dup live.
32      - **#28 --encoder nvenc** (proxy + batch): GPU h264_nvenc (`-rc vbr -cq <crf> -b:v 0`,
preset p5); libx264 still
33      default. Parity is codec-agnostic. encoder_available() guards a missing GPU. **NVENC
litmus VERIFIED** (RTX 2070S,
34      36901 frames @ 59.94 both); laptop stays cool (GPU encode, I/O-bound). cq19 ? 2x
libx264 size (raise --crf to shrink).
35      - **#29 richer schema** (court+confidence+capture-report+IAA ? owner picked ALL 4): per-
rally `court`
36      (primary|secondary, the multi-court gold label for the 0.66 weakness) + `confidence`
(sure|unsure ? `annotator_unsure`
37      review flag, never penalised) carried Brief+Guide+convert-cvat?canonical CSV?round-
trip; capture-report (per-video
38      quality form, doc-only) + IAA ~15% A/A overlap (Brief $4/$9). ? court/confidence
currently STOP at the canonical
39      CSV+review sheet ? **DB/export integration is a tracked FOLLOW-UP before training on
them** (import-local reads core
40      cols only).
41      - **THE BATCH:** `prep-annotation-batch --encoder nvenc` running on F:\GoPro?\Roora June
15 (12 unique CFR videos,
42      ~115GB 4K/5.3K/one 120fps/one 1080p) ? mirrored proxies-only tree at `G:\My
Drive\Sharing\Annotation Videos\Roora
43      Request - June 15` (Google Drive ? shareable). Manifest (private) on
F:\?\roora_june15_proxy_manifest.csv; batch log
44      F:\?\roora_batch_log.txt (? python stdout block-buffers ? watch G: proxy files +
manifest rows, NOT the log's

```

```

45     print lines). ~3/12 done at checkpoint; ~3.7GB/proxy at cq19 ? ~30-40GB Drive upload
(iterate to higher --crf if slow).
46     - **Roora guidelines Doc** (canonical Google Doc, see [[roora-guidelines-drive-doc]])
now NEEDS regeneration with the
47     new schema ? I can't edit the Doc in place (no Drive write tool); gave owner paste-
ready deltas. ?? owner: fill the
48     Doc TODOs + paste the new sections; A/A duplicate-annotation test deferred (owner
idea: 2 identical proxies ? check
49     byte-identical CVAT return). DB integration of court/confidence = next collector PR
when we train on them.
50
51     **Checkpoint:** 2026-06-14 (? GATE-A REGRESSION RESOLVED + eval?serve FOUNDATION FIXED ?
the "owner decision pending" in the
52     checkpoint below is now DONE). The served Gate-A regression (#146 flip ? ?0.008 served,
drops ~12 real rallies) is **REVERTED**
53     (**#151** ? served default back to `gemini`, `min_crossings=0`, verified) + the served
litmus (**#148**) + finding docs (**#152**)
54     merged. The **foundational eval?serve gap is FIXED** (**#154**):
`backend/eval/windowing.py` single-sources the SERVED preset
55     from production `IndexingConfig` (eval+serve can't drift);
`backend/eval/promotion.py::promote_if_served_safe` VETOES any cue
56     winning on PROPOSAL windowing but regressing SERVED (Gate-A: proposal +0.0739/6-0 ?
served ?0.0077/2-2 ? **promote=False**);
57     `serve_contrast.py` = the worked example. 21 tests, CI green. **[[decision-rigor-norm]]
updated with EVAL?SERVE** (a served check
58     is REQUIRED before flipping any served default). Owner directive "do not regress + fix
foundations before any other task" ?
59     **BOTH DONE.** ?? **OPEN/HELD:** **#149** (exec-policy P3a worker self-scheduling loop,
green, additive) is now UN-BLOCKED but
60     un-merged pending owner go; **#150** (per-user store/bridge v4/v5 + `serving.py`) MERGED
additive/inert (NEXT owner-gated = wire
61     the bridge into `_select_for_handoff`); owned-model Phase-0 PLAN at
`scratch/OWNED_MODEL_PHASE0_PLAN.md` (item 7, for discussion).
62     Repo clean on `master` ? `4251c9c`; worktree `.claude/worktrees/bhi-execpolicy` remains
for #149. **Context ~critical ? keep handoffs granular.**
63
64     **Checkpoint:** 2026-06-14 (SERVED-SIDE GATE-A CHECK ? the #146 flip's offline win does
NOT transfer to the production
65     windowing; mild served regression found, owner decision pending). Ran the owner-gated
follow-up flagged in PR #146: drove the
66     **live `FusionSegmenter` served seams** (`_enrich_candidate_stats`?net_crossings,
`_select_for_handoff`?the gate) over each
67     golden video's **cached** WASB trajectory at the **production windowing**
`process_video` actually uses (vt=0.02, merge_gap=2.0,
68     min_win=2.0 from `load_config()`), no WASB re-run / GPU / Gemini. Repro
`scratch/served_gateA_{verify,sweep}.py`; artifacts
69     `output/gateA_served_verify/`. **a) CONFIRMED:** at `min_crossings=0`, fusion_hybrid
seeding+handoff is **byte-identical to
70     wasb_hybrid on all 6** (no-regression invariant verified on live served code, not just
review). **b) FINDING:** the +0.074
71     win was measured on the eval harness's recall-oriented *proposal* windowing
(0.005,1.0,0.5; CONTROL over-segments seg-ratio
72     1.68?1.01 = Gate-A cleaning fragmentation). The **served coarse windowing already merges
aggressively** ? Gate-A has little FP
73     to remove and instead trims **real rallies** with net_crossings 0?1. Candidate-handoff
F1 (IoU0.5, n=6) sweep: **mc0 0.300 |
74     mc1 0.303 (+0.003, 1 real-rally lost) | mc2(SHIPPED) 0.292 (?0.008, 12 real-rallies
lost) | mc3 0.269 (?0.031, 28 lost)**;
75     worst largetest_doubles ?0.051. Tiny/within-noise BUT mc2 is **not** a served
improvement and drops genuine rallies (dropped
76     windows never reach Gemini = unrecoverable recall loss; precision "gain" partly
duplicates Gemini ? served tradeoff ? neutral).
77     **Root cause = eval?serve windowing** (harness promotes on a windowing production
doesn't serve). **Recorded in
78     `docs/QUALITY_ITERATIONS.md` ? committed as **PR #152 (merged, `53bc743`)**. **?

```

RESOLVED IN PARALLEL by a SIBLING session:**

79 the same regression was independently found + productionized as a standing litmus
 (**#148** `backend/eval/served\_gate\_a.py` +
 80 `tests/test\_served\_gate\_a\_regression.py`) and the **flip was REVERTED in #151**
 (`config.json` default_segmenter ? `gemini`,
 81 `indexing.fusion` block removed; `FusionConfig.min\_crossings` default 2?0 ? Gate A OFF-
 by-default-but-opt-in). My independent
 82 numbers MATCH #151's (?0.008, ~12 rallies dropped, 2/6 losses) = good corroboration. So
 the owner decision I'd queued is M00T
 83 (already reverted). Reconciled my #152 doc entry to point at the revert (#151) + the
 productionized guard (#148) instead of
 84 "decision pending". **OPEN follow-up (real lesson):** align the experiment-harness
 windowing with the served windowing before
 85 treating a harness lift as a served default. Ties [[decision-rigor-norm]] (foundations:
 eval?serve) + [[weak-signals-retained]]
 86 + [[paper-coauthor-aim]]. ? Watch [[gotcha-shared-checkout-branch-collisions]]: this
 whole arc (#148/#151 sibling vs my #152)
 87 was concurrent slates on the same finding ? branch-from-origin protocol held.
 88
 89 **Checkpoint:** 2026-06-14 (PERSONALIZATION Phase-0 LANDED + quality checkpoint).
 Owner adopted the HYBRID ? personalization as a
 90 flag-gated FEATURE on a generalization-first baseline + a **contribution flywheel** ("a
 tool that gets better the more you help it";
 91 free-tier videos pool into the owner-trained baseline). Pivot analysis
 (`scratch/PERSONALIZATION\_PIVOT\_ANALYSIS.md`) recommended
 92 hybrid-not-personalization-first; the owner's contribution-flywheel closes the flywheel-
 loss objection. **MERGED this session:**
 93 telemetry wire-in **#140** ; per-user promotion gate **#141** (min-N floor + structural-
 guard exemption); held-out-USER learning
 94 curve **#142** ("<K videos to superior quality" = falsifiable); per-user no-regression
 guard **#143** ; **`docs/PERSONALIZATION\_PLAN.md`
 95 #144** (the checked-in project doc + locked decisions). Gemini-training-source concern =
 CLEAN (eval-only, gitignored, M0.3 owner-gated).
 96 **LOCKED owner decisions:** identity = north-star-design/current-impl (nullable user_id,
 device UUID); inline classifier_json behind a
 97 cloud-load interface; **`min\_n` is a PROMOTION-confidence floor, NOT a service gate** (a
 1-video user is always fully segmented by the
 98 baseline); free-tier video pooling (consent + minors-exclusion); **Gate-A-only
 structural + `min\_n=5`** . ? **GATE-A MILESTONE DONE:** Gate-A FLIPPED to the served default
 (**#146** , config-reversible ? `default\_segmenter`
 99 gemini?`fusion\_hybrid` + `min\_crossings` 0?2; rollback = one flag); the **ablation
 layer** built + **LITMUS PASSED** (**#147** ?
 100 `backend/eval/ablation.py` LOSO/LOGO runner reproduces Gate-A +0.074 / 6-of-6 / p=0.031
 EXACTLY; PDF export ? also in
 101 `OneDrive\Documents\Rally\_Ablation\_Report.pdf`); quality-iterations checkpoint
 (**#145**); **[[decision-rigor-norm]]** codified
 102 (NUMBERS INFORM, FOUNDATIONS DECIDE). Numbers: Gate-A = the ONLY lever clearing the
 honest bar; audio +0.079 = suggestive lead (CI
 103 spans 0); Gate-B inert/trap (court-mask dep). **GPU + Gemini AUTHORIZED for evals**
 (cooler boost on). ?? **OPEN:** served
 104 `fusion\_hybrid` path is **mock-tested-only on real video** ? a **served golden
 regression** is recommended (re-score cached WASB ?
 105 fusion-default vs golden labels); `reportlab`/`pypdf` ? requirements (CI-exercise the
 PDF). NEXT
 106 personalization = Phase 1 (serving bridge: wire `LogisticRegression.load` into
 `\_select\_for\_handoff`) ? owner-gated, see [[weak-signals-retained]] +
 `docs/PERSONALIZATION\_PLAN.md`.
 107
 108 **Checkpoint:** 2026-06-14 (FUSION P3/P4 MEASURED + extraction COMPLETE + telemetry #134
 merged + Cooler Boost). **THE NUMBERS
 109 (honest LOVO n=6, IOU?0.5, C1 bootstrap-CI + sign-test):** shuttle-only F1=**0.307**;
 +person (P3) ?F1=?0.021 (?6.7%, CI
 110 [?0.165,+0.162], sign 2W/4L p=0.688, **CI-clears-0: NO**) ? NO lift, stays default-OFF
 (no court polygons ? `players\_in\_court`
 111 counts spectators; per-video wash: testlarge +0.40 / gbaaddy ?0.22). **+person+audio (P4

```

incremental) ?F1=+0.079** (+27.7%, F1
112 0.286?***0.366**, CI [?0.054,+0.203], sign 4W/2L p=0.688, **CI-clears-0: NO**) ?
PROMISING (wins 4/6, largetest/mahadevpura
113 +0.27; endpoint 0.366 > shuttle-only 0.307) but n=6 can't confirm. **VERDICT: neither
promotable (both CIs span 0); audio is
114 the lead ? need more matches + measure audio on shuttle-only directly (drop the person
handicap).** Recorded in
115 `eval_baselines/experiment_leaderboard.json`; honest negative/uncertain results KEPT
([[paper-coauthor-aim]]). **LANDED THIS
116 SESSION:** result logged to `docs/QUALITY_ITERATIONS.md` (#137);
`experiment_leaderboard.json` tracked (#139); **weak-signals
117 audit *** (person/audio/court-bounds all present, default-OFF, no-special-casing,
retained ? refuted=false ? [[weak-signals-retained]]);
118 7-page paper-style **quality report PDF** at
`OneDrive\Documents\Rally_Detection_Quality_Report.pdf` (gen `scratch/quality_report.py`;
119 conservative academic restyle, strict-reviewer-approved); `FusionGoldenExtract`
scheduled task **unregistered** (cleanup); tree
120 clean on `master`. ?? **NEXT (fusion track):** (1) **wire `run_telemetry` (#134) into
fusion_golden + the ps1 wrapper** (deferred,
121 ready, mergeable); (2) **measure audio on shuttle-only directly** (quick eval, reuses
the 6 `output/*_golden_features.json`, NO new
122 data ? audio's standalone lift without the person handicap); (3) **grow the golden
corpus via Roora** (owner's job ? gives power
123 to confirm the +0.079 audio lead) + **activate court polygons** (re-measure person
properly ? the ?0.021 is partly a
124 `court_polygon=None` artifact); (4) future **shuttle-dropped** cue ([[weak-signals-
retained]]). **OTHER track (per
125 [[session-handoff]] ? the stated IMMEDIATE PICKUP): EXECUTION-POLICY P3 then P4**
(owner-gated; #132/#133/#135 done). **Extraction infra** (all working): detached Scheduled Task
`FusionGoldenExtract`
126 runs `scratch/extract_until_done.ps1` (self-restart loop + in-process keep-awake + GPU-
busy gate: waits if free VRAM<3500MB) ?
127 deaths were `0xC0000013A` console-control kills (NOT crash/sleep/TDR/OOM ? verified via
event log), auto-recover via 3-min
128 repetition trigger + restart-on-failure. **Cooler Boost** (MSI Creator Center) dropped
85?67°C, SM ~1100?~1300MHz, velocity
129 ~26?~50 fps (the rest is Max-Q power cap, not thermal).
130
131 **Checkpoint:** 2026-06-14 (EXECUTION-POLICY framework: design + P1 + P2 SHIPPED ? **PR
#132 (design) + #133 (P1) + #135 (P2)
132 MERGED**, self-merged on green CI). Owner asked to harden the GX010125 Gemini hang as a
**policy-driven, per-job,
133 self-scheduling** system (default WAIT_AND_RESUME, ABORT, POLL_EVERY_N_SEC?, **no master
node**) with **very high coverage**
134 (failures cause delays) and **clean poll-logging** (visible but not polluting). **#132**
= design doc `docs/EXECUTION_POLICY.md`
135 (validated the owner's architecture: per-job policy = data on the job; self-schedule via
re-enqueue; no master ? rides the
136 existing single-worker executor + `jobs` table + crash sweep; DB is the coordinator).
**#133 (P1, the actual fix)** =
137 `backend/infrastructure/execution_policy.py`: `ExecutionPolicy` + `run_bounded` (HARD
`op_timeout_sec` per-attempt ? the
138 hang-killer; retries failures AND hangs; hung attempt's daemon thread abandoned) +
`poll_until` (bounded poll loop). Clocks
139 injected ? **12 tests, 100% module coverage, instant**. Refactored `gemini.py`:
`generate_content`?`run_bounded` (a
140 never-returning generate previously blocked FOREVER = the GX010125 hang; now bounded),
processing-wait?`poll_until` (**per-poll
141 logs at DEBUG, not an INFO line every 10s** ? the no-pollution ask). 23 gemini tests
stay green. **#135 (P2, substrate)** =
142 DB migration **v3** (`jobs.policy` JSON + `next_poll_at`; existing rows survive via
ALTER) + `set_job_waiting(delay_sec)`
143 (self-schedule: park 'waiting' until due) + **`claim_due_job()`** (atomic compare-and-
set claim of queued/due-waiting ?
144 running; concurrent-safe, no master) + `create_job(policy=)`/`get_job` policy.

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**Additive, no executor change**; 10 tests;
145     version asserts bumped 223. ? **THIS IS THE OWNER'S REAL GPU MACHINE**
(torch+CUDA/cv2/WSL/Gemini live). ?? **REMAINING
146     (owner-gated, NOT done ? deliberately not rushed at session tail given the high-coverage
bar): P3** = wire the executor
147     (main.py) to a `claim_due_job` self-scheduling loop honoring WAIT_AND_RESUME
(release+reschedule) + **resumable AI handoff**
148     (persist per-window confirmation; resume only unconfirmed ? the GX010125 reel becomes
exhaustive across resumes, no wasted
149     Gemini spend); **P4** = apply the policy to the WSL/WASB GPU pass + ffmpeg. Each = its
own distinct PR. Ties [[paper-coauthor-aim]] indirectly (infra).
150
151     **Checkpoint:** 2026-06-14 (ACTED ON FIRST-A/B FINDINGS: Gate-A PROMOTED + learned FIXED
+ GX010125 run ? **PR #130 + #131
152     MERGED** to master `2e5892e`; self-merged on green CI). ? **THIS IS THE OWNER'S REAL
GPU MACHINE** (torch+CUDA, cv2, WSL
153     Ubuntu, Gemini key in .env all present) ? not the dev container the older CLAUDE.md note
assumes. **#130 = learned-variant
154     FIX**: additive/default-OFF `Variant.tune_threshold` (operating point on the TRAIN fold)
+ `calibrate` (Platt/isotonic, C2);
155     validated on n=6 golden ? **mean held-out F1 0.307?0.423 (+0.116), testlarge 0.000?0.300
(un-zeroed), now BEATS CONTROL
156     0.388**; threshold-in-LOVO is the big lever. **#131 = Gate-A PROMOTED** (first cue
promoted via the harness; doc trail in
157     QUALITY_ITERATIONS ? measured win dF1 +0.074, 6/6, p=0.031; recommended
`indexing.fusion.min_crossings=2`; **live served-default
158     flip stays OWNER-GATED** until a real fusion-path run confirms ? fusion P2 is unverified
on real video) **+ new
159     `docs/PER_VIDEO_OUTPUT_LAYOUT.md`** (one dir per video; fixes the hardcoded `output`
joins in `trajectory_hybrid.py:~176`
160     so `--output-dir` truly isolates). **GX010125_1080p RUN (offline, no GPU):** ? key
finding ? `wasb_runner.run_predict` has
161     **NO cache-hit early-return**, so the full pipeline would RE-RUN WASB (~tens of min on
1080p) + overwrite the cached CSV; the
162     cached trajectory is reusable ONLY via offline re-score. So ran
`scratch/run_gx010125_rescore.py` = parse cached
163     `output/wasb/GX010125_1080p__89605e80ed01_wasb.csv` ? 45 candidates ? CONTROL 45 rallies
(11.0min) vs **Gate-A 27** (9.3min),
164     Gate-A **dropped 18 low-crossing windows (~1.7min)** of likely held-shuttle/feed FPs ?
isolated `output/GX010125_run/` (no
165     pollution; no golden labels for this footage ? divergence not F1). A FULL Gemini-
confirmed highlight reel = a separate
166     ~tens-of-min GPU+API run (owner kicks off). Repro:
`scratch/{run_gx010125_rescore,ab_n6_lovo,ab_eval_demo}.py`. All in
167     isolated worktrees off master (siblings concurrent). ?? NEXT (owner-gated): real fusion-
path run to confirm Gate-A served-side
168     + flip the live default; impl `PER_VIDEO_OUTPUT_LAYOUT` L1; Tier-2/3 §13 corrections;
analyzers/reconcilers A1?A5/R1?R3. Ties [[paper-coauthor-aim]].
169
170     **Checkpoint:** 2026-06-14 (FIRST GOLDEN A/B EXPERIMENT + HARNESS HONEST-REPORTING WIRED
? **PR #128 + #129 MERGED**
171     to master `f680b8d`, self-merged on green CI per [[working-agreement-merging]]; owner-
directed momentum). Ran the
172     **first real end-to-end A/B on the n=6 golden corpus, 100% OFFLINE** (re-scored
persisted WASB trajectories via
173     `distill_local.build_candidates`, no GPU ? the "persist once re-score" thesis proven).
One candidate set re-scored
174     three ways: **Gate-A (net_crossings?2) is a MEASURED WIN ? ?F1 +0.074, 6/6 matches,
sign-test p=0.031, CI
175     [+0.042,+0.104], seg-count-ratio 1.68?1.01** (over-seg nearly eliminated); the **window-
level 5-feature logistic did
176     NOT clear the bar** (?0.081, 2W/4L; testlarge zeroed = threshold/calibration artifact ?
C2 motivation; **distinct from
177     the stronger per-frame `rally_seq_proto` 0.583** ? don't conflate). Headline: **the
framework held up ? C1 promoted one

```

```

178     variant and refused another (right call both ways); C4 surfaced over-seg tIoU hid.**
**#128** = archived posterity doc
179     `docs/archives/quality-iterations/2026-06-first-ab-experiment/` (full tables) +
archives-index + live QUALITY_ITERATIONS
180     pointer. **#129 = (b) wired C1+C4 into the CORE `experiment.compare()`**
(additive/default-ON `honest_stats=True`:
181     `honest` block = per-metric bootstrap CIs + paired ?F1 CI + exact sign test + F1@k +
segment_count_ratio; `record_experiment`
182     carries honest ?F1; existing fields byte-unchanged, `honest_stats=False` = legacy
shape). Reuses #124 helpers; complements
183     sibling **#127** which wired the same C1 into `fusion_compare` (no dup). 19 existing
experiment tests pass + 3 new;
184     ruff+mypy+CI all green. Repro `scratch/ab_n6_lovo.py` + `scratch/ab_eval_demo.py`
(machine-side; read gitignored
185     manifest/external golden). All done in **isolated worktrees off master** (siblings
merged #121-#127 concurrently). ?? NEXT
186     (owner-gated): promote Gate-A via the no-regression gate (it's a real measured win); fix
the logistic (C2 calibration +
187     threshold-inside-LOVO); Tier-2/3 $13 corrections; analyzers/reconcilers impl A1?A5/R1?R3
(none started). Ties [[paper-coauthor-aim]].
188
189     **Checkpoint:** 2026-06-14 (FUSION P3/P4 MEASUREMENT BUILT + GPU EXTRACTION RUNNING ?
master `d7b1f51`; suite 514 green).
190     The "modeling" track for the person-cue (P3) + audio (P4) fusion lift ? all default-OFF,
no-special-casing, harness-measured.
191     **Merged this run (mine): #121** fusion P3 (`backend/eval/fusion_golden.py` offline
person-feature EXTRACTOR + `fusion_compare.py`
192     LOVO driver ? reuses build_candidates/PersonDetector/fusion_features/experiment.compare,
NO new scoring); **#122** P4
193     (`backend/pipeline/detectors/audio_features.py` + `backend/eval/fusion_audio.py`
incremental audio cue); **#123** progress-logging
194     + resumability (`--fresh`); **#125** single-pass
`PersonDetector.detections_over_windows` (fixed the per-window-peek 0(N)
195     pathology ? ~28s/window ? one sequential decode); **#126** `--smallest-first`; **#127**
honest ?F1 = bootstrap CI + paired
196     sign test on per-video held-out F1 (wires #124's `eval_stats.paired_delta_ci`; "never a
bare mean"). Concurrent sessions
197     merged #119 (analyzers framework), #120 (run.log), #124 (eval-honesty utils:
eval_stats/score_calibration/segmentation_metrics).
198     **? GPU EXTRACTION via DETACHED SCHEDULED TASK `FusionGoldenExtract`** (runs
`scratch/extract_until_done.ps1` ? Task Scheduler
199     owns it, OUTSIDE Claude's job object ? survives turn/context/session boundaries; in-
process keep-awake + self-restart loop until
200     6/6; clean UTF-8 logs to `output/fusion_golden.log` + `scratch/extract_wrapper.log`).
Loops `python -u -m backend.eval.fusion_golden
201     --manifest output/human_lovo_manifest.json --out-dir output --smallest-first` ? 6
`output/<name>_golden_features.json` (GITIGNORED
202     + RESUMABLE: skips any video whose JSON exists). **3/6 DONE** (testlarge, mahadevpura,
largetest); remaining
203     kushagra?gbaaddy?adarsh(last, 84k frames). **MONITOR:** JSON count + `tail
output/fusion_golden.log` + `nvidia-smi`;
204     `Get-ScheduledTask FusionGoldenExtract|Select State`. **RESTART (resumes, skips done):**
`Start-ScheduledTask -TaskName
205     FusionGoldenExtract`. **STOP:** `Stop-ScheduledTask -TaskName FusionGoldenExtract` +
`Get-Process python|Stop-Process -Force`.
206     ?? **WHY DETACHED (debugged 2026-06-14):** Claude-bg-task runs DIED TWICE silently
(largetest 61%, kushagra 6% @137s), NO Python
207     traceback AND ? verified via Windows event log ? NO sleep (Kernel-Power 42), NO TDR
(Display 4101), NO WER crash, NO RADAR-OOM
208     (no event 1003; 18GB free). Traceless vanish = external `TerminateProcess`; common
factor = both were Claude-managed bg tasks
209     (job-object kill-on-close at a context boundary fits death #1 @ the prior context
summary). Scheduled Task = immune; GPU access
210     CONFIRMED in the interactive task session (53?64% util, 3.1GB). GPU util while running =
20?64% bursty ? batch=1 Faster-RCNN

```

211 latency-bound (small kernels + CPU NMS + per-frame ByteTrack), NORMAL, model IS on CUDA, NOT decode-bound; batching the forward

212 pass is the next-extraction speed lever (changes features, so NOT mid-run). ~1.5-2h total. Stale Claude bg keep-awakes stopped. ****DEATH #3 + GPU-job check (2026-06-14):**** died a 3RD time (gbaaddy 69%)

213 ? `LastTaskResult=0xC000013A` = STATUS_CONTROL_C_EXIT = terminated by a ****console-control event / CTRL+C**** (NOT crash/sleep/TDR/OOM

214 ? explains the no-trace vanish; killed the detached task too, so NOT Claude's job object). GPU-contention angle CHECKED: ****NO**

215 competing GPU compute job****** (only desktop apps + our 1 python; 5GB free; HW is a ****laptop**** RTX 2070S ****Max-Q****, MSI). Mitigations:

216 (1) task re-registered with ****3-min repetition trigger + restart-on-failure (count 10, 1-min)**** ? auto-recovers within minutes of

217 ANY death; (2) ****GPU-busy gate in the ps1**** (owner policy = **wait until free, auto-resume**): before each python launch, if free

218 VRAM <3500MB (another job has the GPU) it waits 30s + re-checks, then launches. ****??**

WHEN IT COMPLETES ? the remaining steps for

219 the NUMBERS:****** (1) `python -m backend.eval.fusion\_compare --features-dir output --record` ? honest P3 LOVO ?F1 (shuttle-only

220 vs +person) w/ CI+sign-test; (2) `python -m backend.eval.fusion\_audio --augment --compare --record` ? P4 incremental audio

221 ?F1; (3) log both to `docs/QUALITY\_ITERATIONS.md` I3 (HONEST measured numbers only ? [[paper-coauthor-aim]]). Honest caveat:

222 NO court polygons ? `players\_in\_court` counts spectators (3-7 on "single-court" Mahadevpura); robust signals = motion +

223 `shuttle\_player\_colocation`. GPU = RTX 2070S, person detection on Windows torch cu124 (NO WSL). Branch protocol held through

224 heavy concurrent merges (#119-127) ? [[gotcha-shared-checkout-branch-collisions]]. On master, clean.

225 ****STORAGE** (this session, owner-approved "delete cache + compact WSL"): ****** deleted the stale ****93GB GX010125 frame cache****

226 (WSL `~/clips` 95?2.3GB; WSL fs used 152?60GB, 896GB free; C: 53?72GB free) ? kept the 2.2GB GX010125 proxy. The WSL

227 `ext4.vhdx` was `--set-sparse true` + `fstrim`'d (946GB trimmed) but this WSL build (2.3.24) did NOT auto-deallocate on

228 shutdown ? the file is still ****~221GB** allocated on C:******. ****DEFERRED** to post-extraction: the physical ~150GB C: reclaim via

229 `wsl --shutdown` then elevated `diskpart` ? `select vdisk file="...\CanonicalGroupLimited.Ubuntu\_79rhkpf1ndgsc\LocalState\ext4.vhdx"`

230 ? `compact vdisk`****** (heavy C: I/O would fight the GPU run's video reads). [[gotcha-long-runs-gpu-wsl]] · [[cache-hygiene-policy]].

231

232 ****Checkpoint:**** 2026-06-14 (RALLY-DETECTION FRAMEWORK doc + LITERATURE REVIEW + eval-honesty code ? ****PR #119 + #124 MERGED****

233 to master via self-merge on green CI per [[working-agreement-merging]]; owner-directed). ****PR #119**** (`docs/ANALYZERS\_AND\_RECONCILERS.md`):

234 reframed the analyzers/reconcilers design into a ****multi-scope signal-fusion framework**** (frame detectors / window analyzers /

235 ****session analyzers****; warm-up `SessionAnalyzer` + shuttle-state frame-detector folded in) + ****§13 = a verified multi-agent**

236 literature-review****** (8 patterns surveyed, ~40 citations adversarially checked, ****ZERO fabrications****; author-line fixes applied).

237 ****Verdict (honest):**** the skeleton is STANDARD prior art ? candidate-windows+scorer = Temporal Action Localization;

238 reconciler-as-refinement = Temporal Action Segmentation post-proc; "signal not verdict" = late/score-level fusion + stacked

239 generalization (~20 yrs old, already in racket sports). ****Only the report-first reconciler-as-linter bundle rated**

240 `novel-combination`******; warm-up-as-soft-feature = thinnest gap. ****Cite, don't claim****; terminology realigned (producer?labeling

241 function, scorer?stacked meta-learner, LOVO?leave-one-group-out, tIoU?F1@k/mAP). Ties [[paper-coauthor-aim]] (§13 is the honest

242 related-work backbone). ****PR #124**** implements the ****Tier-1 measurement-honesty course corrections C1?C4**** as PURE ADDITIVE /

243 default-OFF helpers (no existing compare/LOVO/metrics behaviour changed):

```

`backend/eval/eval_stats.py` (C1 `mean_with_ci`/
244   `bootstrap_ci`/`paired_sign_test`/`paired_delta_ci` ? never a bare LOVO mean at n?6; C3
`out_of_fold_predictions`/`grouped_folds`
245   ? no stacking leakage), `score_calibration.py` (C2 Platt+isotonic+brier/ECE ? comparable
cue confidences), `segmentation_metrics.py`
246   (C4 F1@{10,25,50} + mAP@tIoU + segment_count_ratio ? make over-segmentation tIoU is
blind to visible). 24 tests; ruff+mypy clean;
247   all 4 CI lanes green (full-suite coverage floor held). Both PRs done in **isolated git
worktrees** (off master) to dodge the
248   recurring shared-checkout branch-flips (siblings merged #117 fusion-P2, #118, #120, #123
concurrently; **#116 closed ? its content
249   had landed via #117**). ?? NEXT (owner-gated): **wire C1?C4 into the DEFAULT harness
reporting** (`compare`/`run_regression.py`
250   emit CIs + F1@k/edit-score instead of a bare mean ? changes reported numbers, so its own
PR) + the Tier-2/3 corrections in §13
251   (global reconciler decode, cue-correlation modeling, temporal smoothing, external
ShuttleSet eval). Analyzers/reconcilers IMPL
252   (A1?A5/R1?R3) still owner-gated, none started.
253
254   **Checkpoint:** 2026-06-14 (RUN.LOG READABILITY for manual QA ? **PR #120 MERGED ?
master `6076eef`**); owner asked ? self-merged
255   on verified-green CI per [[working-agreement-merging]] (squash; remote+local branch
deleted); was branch `feat/runlog-readability`
256   off `35a50f3`; CODE, minimal/config-gated). Owner's manual-QA loop = run the tool on a
video ?
257   eyeball the rally CSV against `run.log`. Two readability changes: (1) **suppress HTTP
chatter** ? new central helper
258   `backend/utils/logging_setup.py::setup_logging(verbose_http=False)` raises
`httpx/httpcore/urllib3/google_genai/google_genai/
259   google.auth/grpc` to WARNING (NOTSET to restore), gated by NEW knob
`logging.verbose_http: false` (`LoggingConfig` on
260   `AppConfig`); our `backend.*` loggers untouched; wired into `run_process_and_stitch.py`
(at import + re-applied
261   post-config-load). (2) **end-of-run rally summary** ? new pure helper
`backend/tools/export.py::format_rally_summary` (next to
262   `segments_to_csv`/`basic_stats`): header (total rallies + total play HH:MM:SS) + table
`# start end dur(s) shot_count` from the
263   produced play_segments so log?CSV line up row-for-row; handles metadata dict-or-JSON-
string, missing shot_count?`-`; HH:MM:SS
264   via local `_fmt_hms` matching the segmenters' `_fmt_ts`; logged as ONE block at end of
`main()` (also on the all-filtered
265   early-return). Tests: `test_logging_setup.py`, `test_rally_summary.py`, + `verbose_http`
default/override in `test_config.py`.
266   Suite **477 green**, ruff (`.`) + mypy (`backend training`) clean. ? Commit-msg first
attempt mangled (PowerShell `@'?'@`
267   here-string in the *Bash* tool ? literal `@`; amended via `-F file`) ? Bash tool ?
PowerShell. **BRANCH-COLLISION CLEANUP
268   (owner asked): **after merge, LOCAL checkout left clean ? on
`master`==`origin/master`==`6076eef`, 0 ahead/behind; pruned 6 stale
269   LOCAL branches mapped to MERGED/CLOSED PRs (#92/#104/#111/#106/#102 + #116-closed-
redundant) + the stale remote-tracking refs;
270   **kept** `docs/analyzers-framework` (#119 OPEN). Working tree = only the intentional
untracked carryovers (`scratch/`,
271   `artifacts/`). REMOTE still has 15 prunable branches (14 merged-PR + #116) ? **offered
to prune, owner did NOT answer the
272   question ? left untouched** (must KEEP `feat/khelacharya-shot-intelligence` #11 =
deliberately-retained sibling layer per
273   [[sibling-internship-repo]], + #119). ?? NEXT: owner does the machine-side check (run on
a real video ? eyeball the quieter log +
274   summary block; dev container can't run the GPU pipeline); remote-branch prune available
on request.
275
276   **Checkpoint:** 2026-06-14 (ANALYZERS DOC ? FRAMEWORK + paper framing ? **PR #119 OPEN,
owner-gated, do NOT merge**); branch
277   `docs/analyzers-framework` off master `35a50f3`; docs-only). Owner fed two more ideas

```

this session and asked to fold them in

278 and treat the doc as a ****reusable framework / research paper****, not a project patch
(ties [[paper-coauthor-aim]]). Extended

279 the on-master `docs/ANALYZERS\_AND\_RECONCILERS.md` (which had landed via #117) from "two
seams" into a ****multi-scope****

280 **signal-fusion framework**: producers emit SIGNALS at ****3 temporal scopes**** ? frame
detectors (`RallyCue`) / window analyzers

281 / ****session analyzers**** ? all consumed as feature columns by the learned scorer (spine
unchanged: ****no special-casing****, signal

282 ?scorer, never `if/else`). Folded in: (1) ****warm-up/practice**** as a new
****SessionAnalyzer**** tier (whole-timeline span ? an

283 `in\_warmup` membership feature the scorer DOWN-WEIGHTS, NOT a hard `if t<warmup_end:
drop`; measurable on golden TODAY since

284 warm-up is already labeled negative); (2) ****shuttle-state****
(`in\_air`/`racket\_contact`/`inactive`) placed correctly as a

285 ****frame-scope detector**** (a MULTI_SIGNAL_FUSION_PLAN `RallyCue`, NOT a window analyzer)
? derive-from-existing-signals first

286 (velocity+visibility+direction-reversals+audio thwack), dedicated per-frame model ONLY
where WASB trajectory gaps bite, with

287 the per-frame-label blocker flagged. Added ****\$10 generality & research positioning****
(weakly-supervised temporal segmentation;

288 standard-vs-contributed) with an explicit ****attribution honesty caveat****: NO fabricated
citations/results, related-work +

289 ablations + >1-sport still owed, n=6 = direction-check only ? the honest backbone the
paper aim needs. Diff = `2-seam?framework`

290 (310+/137?). ? Shared checkout had flipped to master mid-task (the sibling's collision
lesson recurred) ? re-verified branch,

291 branched fresh, normal commit (no in-flight WIP left to isolate; #117/#118 already
merged so all cross-links resolve). PR #116

292 stays CLOSED (content on master via #117). ?? NEXT (owner-gated): review #119; impl
tracks unchanged (fusion P3 LOVO, then

293 analyzers/reconcilers A1?A5 / R1?R3 per the doc's phasing ? each ONE PR, default OFF,
promote on measured LOVO lift).

294

295 ****Checkpoint**** 2026-06-14 (FUSION P2 SHIPPED + collision cleanup + paper aim ? master
`35a50f3`; suite 467 green; 0 open PRs).

296 ****PR #117 MERGED** = `FusionSegmenter` (fusion P2, registry `fusion\_hybrid`, ****default-
OFF****): shuttle-primary (windows seed

297 from WASB/TrackNet) + adjudicating cues ? `net\_crossings` (Gate A, always persisted,
GPU-free via rally_gate w/ a reused

298 axis estimate) + optional person cue (Gate B) persisting the ~5 `fusion\_features.py`
features (pure-tested) via

299 `PersonDetector.detections\_per\_frame` over each window's ORIGINAL-video frame range.
Plugs in via ****two IDENTITY hooks**** on

300 `trajectory\_hybrid` (`enrich\_candidate\_stats`/`select\_for\_handoff`) ? wasb/tracknet
byte-identical (adversarially verified).

301 Served gating optional+default-OFF (`fusion.min\_crossings`/`min\_players`) ? gates the
handoff while persisting the full

302 SUPERSET (offline substrate intact). Court mask = optional `fusion.court\_polygon`.
****Design-first**** (`fusion-p2-design`

303 workflow) ? ****31-agent adversarial review**** (25 findings?19 confirmed; headline = NO-
REGRESSION verified) ? fixes folded

304 (frame-dim guards, round() frame idx, rally_gate `estimate` reuse, honest torch
docstrings, +6 tests). ****PR #118 MERGED**** =

305 README index fix (QUALITY_ITERATIONS + EXPERIMENT_HARNESS P1-SHIPPED). ****? SHARED-
CHECKOUT COLLISION (lesson)**** the

306 spawn_task DESIGN session ran in THIS checkout, committed `0f673ff` (the analyzers
design doc) + flipped the branch BETWEEN

307 my branch-point check and my `checkout -b` ? my fusion branch rode on top ? ****#117's
squash absorbed the design doc****, so

308 ****`docs/ANALYZERS\_AND\_RECONCILERS.md` is on master via #117**** and ****PR #116 was CLOSED
as redundant**** (content intact on

309 master; #116's stale "OPEN do-not-merge" checkpoint below is superseded). Re-verify `git
branch` AND that no sibling commit

310 landed since, before `checkout -b`. ****Owner side-aim ratified: co-author a paper****

```

"local, cheap & efficient rally detection
311   via multiple clever ideas" ? [[paper-coauthor-aim]]; `docs/QUALITY_ITERATIONS.md` is the
ablation-ready empirical backbone
312   (honest measured results only). **Analyzers+reconcilers DESIGN on master (owner-gated,
NOT started):** injectable per-window
313   analyzers (e.g. net-hit service-fault) + a report-first reconciliation pass; spine =
signals?learned-scorer, **no
314   special-casing** (owner directive). ?? NEXT (owner-gated): **fusion P3** (person
features ? `experiment.compare` LOVO lift on
315   the 6 golden videos ? the eager-person-compute cleanup lands here) + measure the Gate-A
held-shuttle drop on GPU; then
316   analyzers/reconcilers impl; P4 audio. Earlier this session: PR #114 harness, #115
person-cue extraction (see checkpoints below).
317
318   **Checkpoint:** 2026-06-14 (DESIGN DOC: two extensibility seams ? **PR #116 OPEN, owner-
gated, do NOT merge**; docs-only,
319   branch `docs/analyzers-and-reconcilers` off master `df7d5df`). Produced
`docs/ANALYZERS_AND_RECONCILERS.md` (367 lines, a
320   peer to EXPERIMENT_HARNESS + MULTI_SIGNAL_FUSION_PLAN) + a 1-line `docs/README.md`
cross-link. **Spine (owner directive):
321   NO special-casing in the decision path** ? every analyzer/reconciler emits a SIGNAL
(value+confidence); the SCORING MODEL
322   learns to weight it (already true in miniature: `experiment.predict`/_gate_mask` treat
`net_crossings`/_players_in_court`
323   as feature columns). **Seam 1 = injectable `WindowAnalyzer`s (forward)**: a registry
(mirrors `segmenter_registry`) of pure
324   per-window cues over the already-produced per-frame-aligned signals (WASB trajectory +
`PersonDetector.detections_per_frame`
325   + `rally_gate` net); each emits an `AnalyzerResult` = confidence-scored features ?
`candidate_segments.stats`, events ?
326   `events` table (buffer-on-candidate, flush-on-confirm), optional boundary hints; worked
example = a **service-fault**
327   analyzer (shuttle halts in net region ? `service_fault` signal + `net_hit` event + END
hint); injects at the FusionSegmenter
328   `_enrich_candidate_stats` hook, flag-gated default OFF. **Seam 2 = reconciliation pass
(backward)**: a report-first "linter
329   for rally decisions" ? `ConsistencyRule`s (trailing-dead-time?truncate *(owner's misfire
fix)*, no-net-crossing?drop,
330   over-merged?split, boundary-slack?tighten) flag decision-vs-evidence conflicts;
**report-first/apply-optional, NEVER silent**;
331   expressible as a harness `Variant` transform so LOVO-measurable on the 6 golden videos
BEFORE auto-apply; idempotent, fixed
332   precedence. ? **DESIGNED AGAINST the IN-FLIGHT fusion-P2 WIP** (uncommitted in the
shared tree: `fusion_hybrid.py`,
333   `fusion_features.py`, `FusionConfig.cue_flags`, `person_detector.detections_per_frame`,
`docs/QUALITY_ITERATIONS.md`) ? that
334   WIP was kept **byte-intact** (I isolated my single README line via git plumbing; tree is
CRLF / autocrlf=true, restored
335   owner README to exact session-start bytes 3841B). **Merge-order:** the doc cross-links
QUALITY_ITERATIONS.md + the fusion
336   hook, neither on master yet ? **merge #116 after (or with) fusion-P2** (noted in the PR
body). ?? NEXT: owner reviews #116;
337   implementation is the phased A1?A3 / R1?R3 plan in the doc (each ONE PR off master,
default OFF, promoted only on measured
338   LOVO lift) ? all owner-gated, none started. Ties [[candidate-segments-finetuning]] + the
harness ([[working-agreement-merging]]).
339
340   **Checkpoint:** 2026-06-13 (RALLY-QUALITY TRACK: HARNESS + PERSON-CUE EXTRACTION SHIPPED
? **PR #114 + #115 MERGED**,
341   master `df7d5df`; suite 421?441 green; 0 open PRs). **PR #115 (fusion P1):** extracted
the torchvision person detector
342   + ByteTrack + velocity windowing out of `yolo_hybrid` into
`backend/pipeline/detectors/person_detector.py`
343   (`PersonDetector`, a reusable cue producer); `yolo_hybrid` now ORCHESTRATES only
(chunk/prefetch/AI-handoff/DB) and

```

```

344     delegates Stage-1 to `self.person`. **No behavior change** ? proven by the unchanged
`process_video` output tests (mock
345     seam moved to `self.person`); pure detection unit tests ? new
`tests/test_person_detector.py`. person_detector is
346     mypy-CLEAN (`self.model: Any`) so it needed NO quarantine; the torchvision code that put
yolo_hybrid in quarantine left
347     it, but yolo_hybrid STAYS quarantined for PRE-EXISTING orchestration debt (implicit-
Optional process_video defaults,
348     Success|Failure ABC-override mismatch, a Future[...] annotation ? flagged in mypy.ini to
ratchet separately). ruff+mypy
349     clean, CI 4/4. Docs synced (fusion-plan P1?DONE, CODE_MAP person_detector entry +
yolo_hybrid-as-orchestration, NEXT_STEPS
350     item 2 ticked). ?? **NEXT (owner-steered):** **PR 3 = fusion P2.** The cheap, highest-
confidence, PURE-LOGIC piece =
351     **wire Gate A (`min_crossings`) into the live trajectory path**
(`trajectory_hybrid`/`wasb_hybrid`) + add the knob to
352     `IndexingConfig` ? **fixes the owner's held-shuttle false positive*, default OFF,
immediately A/B-able via the harness on
353     the 6 golden trajectories. Gate B (court-mask + player-count) is **owner-gated** on the
court-mask-source decision
354     (MULTI_SIGNAL_FUSION_PLAN §9.1: manual polygon vs auto homography ? owner). **P3 learned
fusion + the SxS-on-new-videos
355     RUN need GPU/owner-side** (persist person features on real video ? train on the 6 golden
? `compare_unlabeled(CONTROL,
356     +person)` on new footage). Two PRs this session within [[working-agreement-merging]]
self-merge authority.
357
358     **Checkpoint:** 2026-06-13 (EXPERIMENT HARNESS P1 SHIPPED ? **PR #114 MERGED**, master
`ecee786`; suite 421?440 green).
359     Picked up the rally-quality track (the desktop-agent pick-up `NEXT_STEPS` flagged; both
design docs `EXPERIMENT_HARNESS.md`
360     + `MULTI_SIGNAL_FUSION_PLAN.md` arrived via the session-start `git pull`
15e4e0e?9231af2, PR #112). Built
361     `backend/eval/experiment.py` = the thin variant/experiment layer from
`EXPERIMENT_HARNESS.md` P1 (the audit's "~80%
362     already exists" HELD ? verified file:line; pure glue over
metrics/calibration/classifier/training/regression.gt_hash/
363     gemini_refine.merge_overlaps/query_candidate_segments, **NO new scoring math**):
`Variant` (JSON recipe:
364     segmenter+config_overrides+cue_flags+Gate-A `min_crossings`+Gate-B `min_players`+learned
classifier; `spec_hash`; pinned
365     `CONTROL`), `compare` (labeled A/B, **LOVO-honest** train-on-others for learned
variants, B?A deltas + `label_set_hash`),
366     **`compare_unlabeled`** (the owner's **SxS-on-NEW-videos** ask ? agreed/a_only/b_only
via match_intervals, no GT),
367     `record_experiment` (? `eval_baselines/experiment_leaderboard.json`),
`load_video_candidates` (DB bridge). 19 tests incl.
368     the load-bearing **no-regression invariant** (all cues OFF ? byte-identical to CONTROL);
ruff+mypy clean; CI 4/4. Docs
369     synced (EXPERIMENT_HARNESS P1?DONE, CODE_MAP entry + stale regression-baseline-path fix,
NEXT_STEPS item 3 ticked). Owner
370     answered ramp-up Qs: **corpus = all 6** golden videos (he said "5";
LargeTestVideo/TestLargeVideo may be cuts of one match
371     ? kept all 6 per the manifest); **merge authority = self-merge on green then continue to
PR 2** (within [[working-agreement-merging]]).
372     ?? NEXT (this session): **PR 2 = fusion P1** person/vision-cue extraction (lift
torchvision+ByteTrack+velocity out of
373     `yolo_hybrid` ? `backend/pipeline/detectors/person_detector.py` producer; yolo_hybrid
consumes it, no behavior change ?
374     regression-safe). Then P3 learned fusion trains on the 6 golden videos +
`compare_unlabeled(CONTROL,+person)` SxS on new
375     footage (GPU/owner-side). Tracks [[candidate-segments-finetuning]] + [[collector-
indexer-bridge]].
376
377     **Checkpoint:** 2026-06-13 (WORKING-TREE CLEANUP ? master `15e4e0e`, tree now CLEAN).

```

```

Cleaned up the leftover
378     uncommitted bits (the cache-hygiene CODE was already MERGED as #109/#110 per the
checkpoint below ? nothing to rescue
379     there): (1) `config.json` GPU-run knobs **reverted to committed defaults** (was
wasb_hybrid/7200/8 ? gemini/1800/1;
380     values stay recorded in the cache-hygiene checkpoint below ? re-apply
`default_segmenter: wasb_hybrid` in one line if
381     that's the intended new default; user didn't answer keep-vs-revert, took the safe
recoverable path). (2) stray root
382     `arial.ttf` (unused ? shipped watermark font is committed
`backend/assets/fonts/RobotoSlab-Regular.ttf` from #103) ?
383     moved to `scratch/`. (3) genuine untracked draft `docs/STORAGE_SHARING_MODEL.md`
(190-line storage-anchored
384     monetization model, owner co-design pending) ? branched + **PR #111 OPEN (owner-gated,
docs-only ? do NOT auto-merge**;
385     supersedes COMMERCIALIZATION.md framing if ratified ? ADR then). Master tree now shows
ONLY intentional carryovers
386     (untracked `scratch/` handoffs+repro scripts, `artifacts/` Gen-0 bundle) ? could be
gitignored for a pristine
387     `git status` if desired (offered, not done). Reminder: `python -m tools.cleanup_caches
--apply` reclaims the 96.5 GB
388     the SessionStart hook flagged.
389
390     **Checkpoint:** 2026-06-13 (CACHE HYGIENE shipped ? **PR #109 + docs #110 MERGED**,
master `15e4e0e`; + the ?5 reel +
391     manual cleanup). **#110** = diligence docs follow-up (README $Cache Hygiene + knobs +
tools/ line; CODE_MAP detector
392     self-clean, tools/cleanup_caches.py entry, common-task rows; min_shot_count API-parity
note) ? docs-only, CI 4/4 green.
393     Both repos now have **0 open PRs**. After the GX010125 E2E run left ~163 GB in WSL
`~/clips`, built a full cache-hygiene system (all on master):
394     (1) **self-cleaning runners** ? `indexing.{wasb,tracknet}.keep_frames` (default false)
deletes the frame cache +
395     staged video after a SUCCESSFUL run; on failure keeps them for resume; (2)
**tools/cleanup_caches.py** dry-run-first
396     ops tool (`--apply`/`--keep-video`/`--older-than`/`--include-wasbcache`/`--summary`);
(3) **disk guard** `min_free_gb`;
397     (4) **SessionStart nudge hook** in `.claude/settings.local.json` (local, gitignored).
Suite 409?**421 green**, ruff+myppy
398     clean. Policy saved ? [[cache-hygiene-policy]]. ? WSL-scan footgun: the owner's login
shell mangles `for e in` loops
399     (loop var empty) ? use du/find arg-globbing. Also this session (earlier): merged
**#107** (API min_shot_count) + **#108**
400     (watermark Windows colon fix); ran GX010125 E2E on GPU ? **GX010125_highlights.mp4** (11
rallies >7 shots) +
401     **GX010125_highlights_min5.mp4** (24 rallies ?5, watermark now works ? validates #108)
via a stitch-only reuse script
402     (no re-inference); confirmed **Gemini got 1080p** (4K only at final stitch); delivered
an execution flowchart. Manual
403     WSL cleanup freed ~68 GB (kept GX010125's 93 GB frames + proxy per owner choice). ?
LOCAL uncommitted `config.json`:
404     default_segmenter=wasb_hybrid / min_shot_count=8 / wasb.timeout_sec=7200 from the run.
See [[gotcha-long-runs-gpu-wsl]].
405
406     **Checkpoint:** 2026-06-13 (SYNC + HANDOFF REFRESH ? read-only reconciliation by a ramp-
up session; no new code from
407     me this pass). `git pull --ff-only` = already up to date at **master `6de70aa` (#108)**.
Reconciled remote vs my last
408     session (#100): seven PRs landed from OTHER slates (NOT my work ? facts from git/gh):
**#101** configurable branding
409     watermark . **#102** thread config stitching paddings into live VideoCompiler sites .
**#103** bundle Apache-2.0
410     default watermark font + fallback log . **#104** C13 field kit (flyer/playbook/answer-
sheet/signals) . **#105** i18n
411     design plan (Kannada-first) . **#107** enforce `stitching.min_shot_count` on

```

```

/api/compile . **#108** escape Windows
412     drive-letter colon in ffmpeg filtergraph (fixes #103 watermark on Windows). **OPEN:
#106** docs/video-locality-model
413     (awaiting owner). ? **UNCOMMITTED WIP IN TREE (sibling/owner ? DO NOT CLOBBER):** a WSL
**cache-hygiene** slice in
414     progress, built on my `wslCommandMixin` ? `base.py` (`_wsl_free_gb`/`_wsl_rm_rf`),
`wasb_runner.py`+`tracknet_runner.py`
415     (`keep_frames`/`min_free_gb` config + pre-run disk guard + post-success cache delete),
`models.py` (new fields),
416     `config.json` (the GPU-run knobs: default_segmenter=wasb_hybrid, min_shot_count=8,
wasb.timeout_sec=7200), untracked
417     `arial.ttf` (watermark font) + `docs/STORAGE_SHARING_MODEL.md`. Not yet PR'd by whoever
owns it. [[session-handoff]]
418     rewritten to this state.
419
420     **Checkpoint:** 2026-06-13 (FIRST FULL E2E GPU RUN on a real 4K GoPro match + 2 merged
fixes).
421     Owner asked to run the indexer E2E on the GPU over `F:\GoPro Hero Black
12\KhelSutra\Badminton 12
422     June 2026\GX010125.MP4` (12.8-min 4K60, 11.5GB), highlights for rallies with >7 shots,
config-driven.
423     - **DELIVERED:** `output\GX010125_highlights.mp4` ? **11 rallies (shot_count 8?16) of 52
detected**;
424         true 4K 3840x2160 h264 59.94fps, 2:23, 1.15GB; stitched from the 4K ORIGINAL (proxy
only for indexing).
425         wasb_hybrid: 64.8% shuttle visibility, 18 candidate windows ? 52 Gemini-confirmed
rallies. min_shot_count=8
426         correctly excluded a 7-shot rally (">7" semantics). Gemini ran clean (no 503s);
~47-min wall clock.
427     - **2 PRs merged on green CI** (owner's [[working-agreement-merging]] standing
authorization):
428         **#107** `/api/compile` now enforces `stitching.min_shot_count` (parity w/
run_process_and_stitch);
429         **#108** sticher `_ff_quote` escapes the Windows drive-letter colon ? fixes the on-
by-default
430         bundled-font watermark (PR #103) silently failing on Windows (`fontfile='C:/...'`
broke ffmpeg
431         drawtext parsing; also fixed absolute logo `movie=` paths). Both with regression
tests; suite 409 green.
432     - **The hard-won run recipe is in [[gotcha-long-runs-gpu-wsl]]:** Modern Standby killed
the first 2
433     overnight attempts (keep-awake fix); cv2 PNG extraction (~38min) is the real
bottleneck not the GPU
434     (ffmpeg pre-extract, count-exact for CFR ? skips it); 1080p-proxy-index / 4K-stitch;
per-phase
435     observability (cache manifest for WASB, handoff dir for Gemini). Reusable scripts in
`scratch/`
436     (run_gx010125.py, keep_awake.ps1, probe_.py).
437     - ? LOCAL UNCOMMITTED `config.json`: default_segmenter=wasb_hybrid, min_shot_count=8,
wasb.timeout_sec=7200
438     (set for this run; revert to gemini/1/1800 if not wanted). Watermark fix NOT applied
to the delivered
439     reel (it ran before #108); a re-stitch would hit the DB segment cache + now-working
watermark (~5min).
440
441     **Checkpoint:** 2026-06-12 (latest ? PAID COMPONENT REMOVED + VIDEO-LOCALITY MODEL; **PR
#104 updated `9c84fea` +
442     PR #106 OPEN `85abe26`, both CI green, both AWAITING OWNER**). Owner spec: (1) remove
ALL associate-pay from the
443     field kit ("something else in mind") ? flyer/playbook now "Engagement terms: discussed
individually"; honesty
444     mechanics reworded ("judged on honest capture, not positive answers"); KEPT: WTP
research questions (study subject)
445     + the ACADEMY's Phase-2 session fee (different party ? flagged, owner may strip too).
(2) Storage/upload questions

```

446 answered from code: uploads ARE stored in full (`security.upload\_dir`, NO retention
policy ? BACKLOG); 10GB CANNOT
447 upload (max_upload_mb=4096 rejects at init; ?90MB sequential parts; hours on Indian
uplinks). (3) NEW
448 `docs/VIDEO\_LOCALITY\_MODEL.md` (#106): central insight = timestamps are the product,
video is dead weight (only the
449 compiler touches original bytes, cuttable where the original lives); locality ladder
L0?L4 (L2 proxy-upload reuses
450 the collector ADR-002 frame-identical contract; L1-direct = client?Gemini, zero bytes
through us; L0 = owned-model

10. assistant (2026-06-15T14:58:30.088Z)

I have enough ground truth now. The current-state confirms: today = rally detect + highlight
reel + play-time summary; the watermark text in config is "Generated by Khelsutra.guru"
(lowercase 's'); minors-never-used, opt-in, local-first are real guardrails; multi-court ~0.66
is the hard case; shot detection / AI coach / multisport / Kannada-Hindi are roadmap. The draft
is largely well-constructed and honest. Now let me do a careful overpromise pass on each claim.

Key honesty concerns I've identified by cross-referencing the draft against ground truth:

1. *****finds every rally***** (line 11, "In one line") ? overpromise. Ground truth: clean single-court ~0.93 F1, multi-court ~0.66. "Every rally" is a completeness guarantee the product does not keep, especially in the multi-court case the doc itself calls the hard problem.
2. *****it finds where each rally starts and ends, on its own***** (line 43) ? the "on its own" is fine, but no honesty caveat that boundaries/detection aren't perfect. Minor; the "not" section partly covers it.
3. *****Pulling the real rallies out of that footage ? automatically ? is exactly what KhelSutra is built to do***** (line 24-25) ? this sits right after describing multi-court busy scenes. Reads as "we solve multi-court today." Ground truth: multi-court is the explicit weakness (~0.66), still being improved. "built to do" is aspirational/mission framing but a prospect will read it as a current capability claim about the hardest case.
4. *****the busy multi-court halls that trip up everything else***** (line 78) ? implies KhelSutra is NOT tripped up by multi-court, i.e. multi-court is solved. Contradicts ground truth and even contradicts the doc's own "not" section (line 128) and FAQ (line 136-137, 150).
5. *****Your videos make _your_ tool sharper... the underlying model gets better at *your* courts***** (line 75-78) ? stated in present tense as if per-user sharpening is live. Ground truth/current-state: the personalization/flywheel is ROADMAP ("auto per-user sharpening" listed under roadmap; per-user store is "persisted/loaded but NOT yet read on the served path"). The flywheel headline is wanted, but it must be framed as the model/direction, not a feature working today. The roadmap callout (line 104-106) helps but this section reads present-tense.
6. ****Watermark text mismatch**** ? draft quotes "Generated by KhelSutra.guru" but config.json has "Generated by Khelsutra.guru" (lowercase s). Since the draft quotes it verbatim in quotation marks, this is an accuracy nit.

Let me verify the personalization "not today" framing once more and check the roadmap section's "AI assistant coach" header for over-implication.

The draft's section header "Where it's going ? toward an ****AI assistant coach****" (line 91) and the honesty callout (line 104) are good. The "not" section (line 126) explicitly says shot detection and AI coach are roadmap. That is honest. The main residual risk is the flywheel section's present tense and the multi-court "solved" implications.

Let me now produce the structured review.

11. user (2026-06-15T14:59:12.048Z)

Structured output provided successfully

12. assistant (2026-06-15T14:59:21.584Z)

I've completed the honesty/overpromise review and returned it via StructuredOutput. The task list items are all from prior work and unrelated to this review, so no task updates are needed.

Summary of what I found in the draft at `C:/Users/avidu/Projects/badminton-highlight-indexer/docs/KHELSUTRA\_PRODUCT\_OVERVIEW.md`:

Three must-fix overpromises, all of which an alpha customer would read as broken promises on first use:

- "finds **every** rally" (line 11) ? a completeness guarantee the ~0.93/~0.66 F1 reality does not keep.
- "the busy multi-court halls that **trip up everything else**" + present-tense flywheel language (lines 75-78) ? implies multi-court is solved AND that per-user sharpening is live today, when both contradict ground truth (multi-court is the explicit weakness/moat; personalization is roadmap, per-user store not yet read on the served path).
- "Pulling the real rallies out of that footage ? automatically ? is exactly what KhelSutra is built to do" (lines 24-25) ? bolded right after describing busy multi-court scenes, reads as a solved-today claim about the hardest case.

Two should-fix items, including a concrete accuracy nit: the draft quotes the watermark as "Generated by KhelSutra.guru" but `config.json:79` has "Generated by Khelsutra.guru" (lowercase 's') ? the doc quotes a verbatim string it doesn't match.

The doc's "What KhelSutra is not" section, guardrail language (opt-in / minors-never-used / local-first / pricing-not-set), the "Read this honestly" roadmap callout, and the accuracy FAQ are all faithful to ground truth and are the parts to keep.